Andrew Fleck

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I study when state estimators become unreliable under imperfect sensing and develop methods to analyze and improve their behavior.

Education

Purdue University - B.S. Biomedical Engineering; Minor in Chemistry 2022

GPA: 3.57/4.0. Senior thesis: ECG waveform reconstruction from scanned images using image processing and filtering. Methods included spatial filtering, morphology-guided masking, and FIR/IIR filtering under noise and distortion.

Publications and Submitted Manuscripts

Aditi Acharya, **Andrew Fleck**†, *Geometry-Induced Contraction Degradation and Stabilization of Learning-Enabled Observers. IEEE CDC 2026, accepted; arXiv:2608.14925;* †*equal contribution.*

Aditi Acharya, **Andrew Fleck**, *When Corruption Looks Normal: Identical Inputs, Incompatible Observer Actions. IEEE/SICE SII 2027, submitted.*

Aditi Acharya, **Andrew Fleck**, *The Ghost in the Observer: Silent Propagation and Re-Exposure of Historical Actions. ACC 2027, submitted.*

Research Experience

Independent Researcher - joint research with Aditi Acharya Dec 2025 - Present

- **Nonlinear observer design:** Co-developed an exact estimation-error operator showing how accurate learned measurements can still produce noncontractive updates; derived the scalar-gain recovery boundary and uncertainty-aware gain projection without retraining. Evaluated recovery in synthetic systems and real tracking data.
- **Observer scheduling under ambiguous sensing:** Co-developed a necessary-and-sufficient condition for when available information permits acceptable observer actions; demonstrated in radar-sounding simulations that identical sensing evidence can require incompatible actions despite bounded, diagnostically nominal observer behavior.
- **Robotic state estimation:** Implemented native EKF replay using public NASA Astrobee recordings; demonstrated persistent estimate differences and altered forecasts after a completed correction intervention in paired experiments with identical inputs and initialization.
- **Estimator validation:** Designed controlled EKF experiments across nominal, geometry-dependent, and intermittent sensing; compared covariance, residuals, observation-derived stability, and realized error. Information loss exposed a gap between covariance-based confidence and realized estimation performance.
- **Geophysical estimation (ongoing):** Developing convex, physics-constrained reconstruction with sparse sensing and time-varying observations; built ingestion and preprocessing pipelines integrating NOAA, NASA, and USGS data.

Research Assistant - Verma Lab, Purdue University Oct 2023 - Dec 2025

PI: Prof. Mohit Verma

- Developed vision-based sensing and probabilistic tracking pipelines for RGB and thermal video, integrating learned perception with filtering and trajectory estimation. Used YOLOv8, ByteTrack, Kalman filtering, Hungarian assignment, and Mahalanobis gating to quantify feeding, drinking, and resting activity in pigs.
- Integrated local measurements with national and international environmental data for PM2.5 prediction using Random Forest, XGBoost, and SHAP; processed satellite, GDAS, and NWS data and filtered sensor noise.
- Applied audio filtering to isolate periods of aggregate aggression; deployed real-time data and inference pipelines on Azure for monitoring and decision support.

Selected presentation: *Spatiotemporal State Estimation and Multi-Object Tracking in Hybrid Biological Systems* (Purdue ABE IRS, 2024). Vision-based tracking and probabilistic state estimation for recovering latent dynamics from noisy video data.

Engineering Experience

Client Systems Engineer - Epic Systems Dec 2022 - Jul 2023

- Led components of distributed clinical infrastructure deployment serving more than one million patients; supported secure Azure/Citrix systems, built PowerShell configuration automation, and designed monitoring and incident-response workflows.

Technical Skills

Estimation and control: Nonlinear observers, EKF, recursive filtering, contraction and stability analysis, inverse imaging, convex optimization.
Programming: Python (NumPy, SciPy, OpenCV), MATLAB, C, R, SQL, JavaScript.
Sensing and ML: RGB/thermal imaging, YOLOv8, ByteTrack, data association, Random Forest, XGBoost, SHAP.
Systems: Scientific data pipelines, distributed systems, Azure, Citrix, PowerShell.